Chapter # 6
Sets & Probability

1. **What are different counting rules?**

I. Factorial

The number of all possible arrangements of n objects is called n – factorial. It is calculated as the product of first n natural numbers.

$n! = n \times (n-1) \times (n-2) \times (n-3) \times \ldots \times 3 \times 2 \times 1$

II. Permutations

If r objects are selected from n objects and the change in order of r selected objects affects the characteristic then the total number of possible permutations are ^nP_r.

$$^nP_r = \frac{n!}{(n-r)!}$$

III. Combinations

If r objects are selected from n objects and the change in order of r selected objects does not affects the characteristic then the total number of possible combinations are ^nC_r.

$$^nC_r = \frac{n!}{r!(n-r)!}$$

2. **Random Experiment**

An experiment with two or more possible unpredictable outcomes under similar conditions is called random experiment. It produces different outcomes on different repetitions.

3. **Trial**

A single performance of an experiment is called trial.

4. **Sample Space**

The set of all possible mutually, exhaustive and complementary outcomes of a random experiment is called sample space. It is denoted by S.

5. **Two coins are tossed. Write down the sample space.**

S = {(HH)(HT)(TH)(TT)} n(S) = 4

6. **Write down the sample space if 3 coins are tossed.**

S = {(HHH)(HTH)(THH)(TTH) (HHT)(HTT)(THT)(TTT)} n(S) = 8

7. What are the total numbers of outcomes if 10 coins are tossed?

$$n(s) = 2^{10} = 1024$$

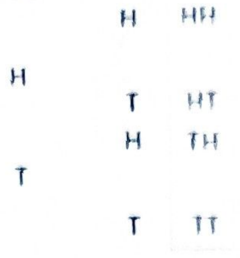
List:

HH HT TH TT

Table:

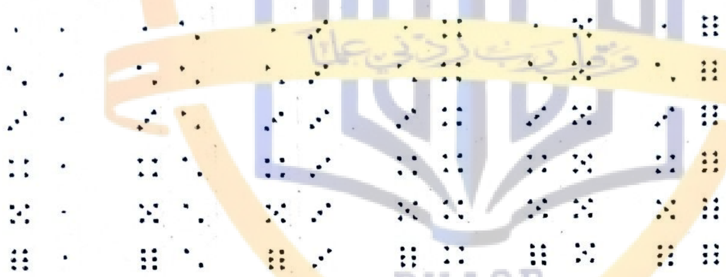
	H	T
H	HH	HT
T	TH	TT

Tree Diagram:



8. Two six faced dice are rolled. Write down the sample space.

$S =$ {(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6),
(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),
(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6),
(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),
(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6),
(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)}; $n(S) = 36$



9. What are the total numbers of outcomes if 3 dice are rolled?

$$n(s) = 6^3 = 216$$

10. What are the total numbers of outcomes if 1 coin and 1 die are tossed together?

$$n(s) = 2 \times 6 = 12$$

Dice

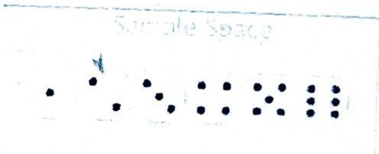
Coin		1	2	3	4	5	6
	H	H1	H2	H3	H4	H5	H6
	T	T1	T2	T3	T4	T5	T6

Important Information about playing cards

Total Cards 52					
Black 26		Red 26		Colours	2
♠Spade 13	♣Club 13	♦Diamond 13	♥Heart 13	Suits	4
♠A	♣A	♦A	♥A	Ace	4
♠2	♣2	♦2	♥2		
♠3	♣3	♦3	♥3		
♠4	♣4	♦4	♥4		
♠5	♣5	♦5	♥5		
♠6	♣6	♦6	♥6		
♠7	♣7	♦7	♥7		
♠8	♣8	♦8	♥8		
♠9	♣9	♦9	♥9		
♠10	♣10	♦10	♥10		
♠ jack	♣ jack	♦jack	♥jack	Jacks 4	Picture Cards 12
♠Queen	♣Queen	♦Queen	♥Queen	Queens 4	
♠King	♣King	♦King	♥King	Kings 4	

11. Sample Point

A single element in sample space is called sample point. It represents a single outcome of a random experiment.



12. Event

A subset of sample space is called event. Event is interested in some specific outcomes that are favouring the occurrence of a concerned typical statement. It is denoted by capital English alphabets A, B, C, etc.

13. Axioms of probability

The probability of occurrence of an event A must satisfy the following properties:

1. Probability of an event cannot be negative.
   i.e., $P(A) \geq 0$
2. Probability of an event ranges from zero to 1.
   i.e., $0 \leq P(A) \leq 1$
3. Probability of sample space is equal to one or the probability of sure event is 1.
   i.e., $P(S) = 1$
4. If A and B are two mutually exclusive events then $P(A \cup B) = P(A) + P(B)$

14. **Classical or prior definition of probability**

If a sample space contains n mutually exclusive, equally likely and exhaustive outcomes of a random experiment i.e., $n(S) = n$, and this sample space contains m outcomes that favours the occurrence of an event A, then the probability of occurrence of event A is denoted by P(A) and calculated as the ratio $\frac{m}{n}$.

i.e., $P(A) = \frac{\text{No.of favourable outcomes}}{\text{No. of all possible outcomes}} = \frac{m}{n}$

15. **Empirical or Relative Frequency or a Posterior definition of probability**

If a random experiment is repeatedly performed sufficiently large n times under uniform conditions and a concerned event A occurred m times then the probability of occurrence of event A is denoted by $P(A)$ and calculated as: $P(A) = \lim_{n\to\infty} \frac{m}{n}$

16. **Mathematical definition of probability**

The ratio between the number of sample points in sample space S and the number of sample points in an event A is called the probability of occurrence of event A and is denoted by $P(A)$.

i.e., $P(A) = \frac{\text{No.of sample points in event A}}{\text{No. of sample points in sample space}} = \frac{n(A)}{n(S)}$

17. **Personalist or Subjective definition of Probability**

The personalist or subjective probability is numerical measure of the confidence or belief regarding the occurrence or non – occurrence of some event based on the experience, intelligence and knowledge of the person.

18. **Impossible Event**

An event containing no sample point is called impossible event. It has zero chances or probability of occurrence.

19. **Sure Event**

An event containing all sample points of sample space is called sure event. It has 100% chances or probability of occurrence.

20. **Simple Event**

An event containing only one sample point is called simple event.

21. **Compound Event**

An event containing two or more sample points is called compound event.

22. Equally likely events

Two or more events are called equally likely events if they have same equal chances of occurrence or non–occurrence.

23. Exhaustive events

Two or more events are called exhaustive events if their union is equal to the parent sample space.

24. Mutually exclusive events

Two or more events are called mutually exclusive events if only one event can occur at a time and they cannot occur together. i.e., the joint occurrence of mutually exclusive events is not possible.

$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$

25. Not Mutually exclusive events

Two or more events are called not mutually exclusive events if two or more of them can occur together. i.e., the joint occurrence of mutually exclusive events is possible.

$P(A \text{ or } B \text{ or both}) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$

26. Independent events

Two or more events are called independent events if occurrence or non–occurrence of one event does not effects the chances of occurrence or non–occurrence of any other event.

$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B)$

27. Dependent events

Two or more events are called dependent events if occurrence or non–occurrence of one event effects the chances of occurrence or non–occurrence of any other event.

$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B/A)$

28. Marginal Probability

The probability of occurrence of an event A regardless the occurrence of any other event is called marginal probability of event A. It is denoted by $P(A)$.

29. Conditional probability

If A and B are two events belonging to same sample space, then probability that an event A occurs given that B has already been occurred is called the conditional probability of A and is denoted by P(A/B).

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

30. Complementary events

Let $\boldsymbol{A}$ be the concerned event belonging to a sample space $\boldsymbol{S}$, then its complement $\bar{A}$ can be obtained by $\bar{A} = \boldsymbol{S} - \boldsymbol{A}$ such that $A \cup \bar{A} = S$.

31. Complementary events rule

Let ***A*** be the concerned event belonging to a sample space ***S***, then its complement $\bar{A}$ can be obtained by $\bar{A} = \boldsymbol{S} - \boldsymbol{A}$ such that $A \cup \bar{A} = S$.

Then $P(A) + P(\bar{A}) = 1$

Where, $P(A)$ = probability of occurrence of event A

$P(\bar{A})$ = probability of non–occurrence of event A

32. Addition law of probability for mutually exclusive events.

If A and B are two mutually exclusive events then the probability either of them happening is$P(A \cup B) = P(A) + P(B)$

33. Addition law of probability for not mutually exclusive events.

If A and B are two mutually exclusive events then the probability that at least one of them happen is $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

34. Multiplication law of probability for independent events.

If A and B are two independent events then the probability both of them occurs is

$$P(A \cap B) = P(A) \times P(B)$$

35. Multiplication law of probability for dependent events.

If A and B are two dependent events then the probability both of them occurs is

$$P(A \cap B) = P(A) \times P(B/A)$$

36. Bayes' Theorem

If A1, A2,...Ak are mutually exclusive and exhaustive events that form a partition of sample space S, and if B is any other event of S, such that it can occur only if one of the Ai occurs then

$$P(A_i/B) = \frac{P(A_i).P(B/A_i)}{\sum_{i=1}^{k} P(A_i).P(B/A_i)}$$

It describes the probability of an event, based on prior knowledge of conditions that might be related to the event.

37. Can two events be mutually exclusive and independent simultaneously?

Two mutually exclusive events A and B are independent if and only if P(A)P(B) = 0, which is true when either P(A) or P(B) = 0.

If both events A and B have non-zero probabilities, they must have a sample point in common. Thus two events that are independent, can never be mutually exclusive.

Moreover, two events are mutually exclusive, are also dependent events, and two events that are not-mutually exclusive, may either be independent or dependent events.

Chapter # 7
Random Variable

1. **Random Variable**

Random variable is the variable whose unpredictable possible values are determined by the outcomes of a random experiment.

Random variables are denoted by capital English alphabets X, Y, Z, etc. and their values are denoted by small English alphabets x, y, z, etc.

2. **Types of Random Variables**

The types of random variable are as follows:

1. Discrete random variable
2. Continuous random variable

3. **Discrete Random Variable**

A random variable which can take only finite number of isolated points in its specified range or interval on the real line is called discrete random variable.

Discrete random is in connection with counting of outcomes.

4. **Continuous Random Variable**

A random variable which can take any number out of infinite possible numbers in its specified range or interval on number line is called continuous random variable.

The values of continuous random variable are typically obtained rny measurements.

5. **Probability mass function**

The function of a discrete random variable taking on distinct values $x_1, x_2, \ldots, x_n$, that describes the total probability associated with each value of X.

6. **Define probability distribution.**

The probability distribution is division of total probability among all the possible values of the variable. It may be expressed in a tabular form by showing all the possible values of X and the associated probabilities.

Values : X	x_1	x_2	x_3	...	x_n
Probabilities : P(x)	$P(x_1)$	$P(x_2)$	$P(x_3)$	...	$P(x_n)$

7. **Properties of discrete probability distribution**

The discrete probability distribution holds the following two properties.

(i) $0 \leq P(x) \leq 1$ which means that the probability lies between 0 and 1

(ii) $\sum P(x) = 1$ which means that the sum of the probabilities is equal to 1

8. Probability density function (p.d.f).

A function defined over the set of all real numbers is called a probability density function of the continuous random variable X if and only if

$$P(a \le X \le b) = \int_a^b f(x)dx \qquad \text{where } a \le b$$

9. Properties of probability density function.

The continuous probability distribution holds the following two properties.

(1) $f(x) \geq 0$

(2) $\int_{-\infty}^{+\infty} f(x)dx = 1$

10. Distribution Function (D.F)

The distribution function of a random variable X, denoted by F(x), is defined by

$F(x) = P(X \le x)$

It gives the probability of the event that X takes a value less than or equal to a specified value x.

It is also called cumulative probability distribution function from smallest up to specific value of x.

11. Properties of Distribution Function.

If $F(X) = P(X \le x)$ is a distribution function of Y then

(i) $F(-\infty) = 0$

(ii) $F(+\infty) = 1$

12. Joint probability Distribution

The probability distribution of two or more random variables, defined on the same sample space S, is called their joint probability distribution.

13. Marginal probability Function

From the joint probability function for (X,Y) we can obtain the individual probability function of X and Y. such individual functions are called marginal probability function.

The values of marginal probabilities are written in the margins of joint table as they are the rows and column totals in the table.

$$g(x) = \sum_{all\ y} p(x,y) \qquad for\ x = x_1, x_2, \dots, x_n$$

$$h(y) = \sum_{all\ x} p(x,y) \qquad for\ y = y_1, y_2, \dots, y_m$$

14. Conditional probability Function

Let X and Y be two discrete r.v's with joint probability function p(x,y), then the conditional probability function for x given Y=y denoted a P(x/y) is defined as

$$P(X = x/Y = y) = \frac{P(x,y)}{h(y)}$$

similarly

$$P(Y = y/X = x) = \frac{P(x,y)}{g(x)}$$

15. Mathematical expectation or Expected value.

The mean or expected value of a random variable 'x' is equal to the sum of the products of each value of 'x' and the corresponding value of P(x) i.e.

$$E(x)=\sum x\, P(x)$$

16. Properties of Expectation

(i) E(c) = c
(ii) E(a + bx) = a + bE(x)
(iii) E(x ± y) = E(x) ± E(y)
(iv) E(x y) = E(x).E(y)

17. Variance of a random variable

The variance of a random variable 'x' is mean of squares minus square of mean.

$$Var(x) = E(x^2)-[E(x)]^2$$

18. Harmonic Mean of random variable.

The reciprocal of expected value of the reciprocal of the random variable is called its harmonic mean. $h = \frac{1}{E(\frac{1}{x})}$

19. Standard deviation

The standard deviation σ of a random variable 'x' is equal to the positive square root of the variance i.e.

Properties of variance	Properties of Standard deviation
For a random variable X	For a random variable X
(i) Var(a) = 0	I. S.D(a) = 0
(ii) V(x±a)=V(x)	II. S.D(x±a)=S.D(x)
(iii) $V(ax)=a^2V(x)$	III. S.D(aX) = |a| S.D(X)
(iv) V(X±Y)=V(X)+V(Y)	S.D(X ±Y)= $\sqrt{V(X)+V(Y)}$

22. **Independence two random variables**

Two discrete random variables are said to be statistically independent if and only if for all possible pair of values (x,y) the joint probability function P(x,y) can be expressed as product of the two marginal probability functions.

$$P(x,y) = g(x).h(y)$$

23. **Covariance of two random variables**

The covariance of 2 r.v's X and Y is a numerical measure of the extent to which their values tend to increase or decrease together.

$$Cov(X,Y) = E[X - E(X)][Y - E(Y)]$$

If X and Y are independent then E(XY) = E(X)E(Y)

It is important to note that covariance is zero when x and Y are independent but its converse is not generally true.

24. **Moment Generating Function (M.G.F)**

The moment generating function (m.g.f) of a random variable x is defined as the expected value of the r.v e^{xt}

i.e. $M_o(t) = E(e^{tX}) = \sum e^{tX}.P(X)$ If X is a discrete variable

i.e. $M_o(t) = E(e^{tX}) = \int_{-\infty}^{+\infty} e^{tX}.f(X)\, d(x)$ If X is a continuous variable

Chapter # 8
Discrete Probability Distribution

1. **Discrete Uniform Probability Mass Function**

The probability mass function of a uniformly distributed discrete random variable X over the discrete set of k values where n is the single parameter of this distribution

$$P(X = x) = p(x) = \frac{1}{n} \quad for \;\; X = x_1, x_2, x_3, \ldots, x_n$$

$$Mean = \frac{n+1}{2}$$

$$Var = \frac{n^2 - 1}{12}$$

2. **Bernoulli trials or Binomial trials**

A series of independent repeated trials of a random experiment having only two possible outcomes, success and failure, are called Bernoulli trials.

3. **Bernoulli Probability Mass Function**

The Bernoulli probability distribution with parameter P is specified by the probability mass function

$$P(X = x) = P(x) = (p)^x (q)^{1-x} \quad for \;\; x = 0, 1$$

$$Mean = p$$

$$Var = pq$$

4. **Binomial Random Experiment**

A series of *'n'* independent trials, repeated with replacement and each having only two possible outcomes, success and failure, is called Binomial Random Experiment.

5. **Properties of Binomial Experiment**

Properties of Binomial Experiment are as follows:

1. The two possible outcomes of a binomial experiment are categorised as success and failure.
2. The experiment is performed a fixed number of *'n'* trials.
3. Probability of success remains same throughout the *'n'* trials.
4. The successive trials are independent.

6. **Binomial Random Variable**

A random variable whose unpredictable possible values are determined by the outcomes of binomial random experiment is called binomial random variable.

7. **Binomial Probability Distribution**

The probability distribution of binomial random variable is called binomial probability distribution.

Its probability mass function is as follows:

$$P(X = x) = {}^{n}C_{x}\, p^{x} q^{n-x} \; ; x = 0, 1, 2, 3, \ldots, n$$

Where, n = number of trials

x = number of success

p = probability of success

$q = 1 - p$ = probability of failure

8. **Properties of Binomial Probability Distribution**

Properties of Binomial probability distribution are as follows:

1. There are two parameters of binomial probability distribution, n and p.
2. The shape of binomial probability distribution is dependent on value of parameter p.
   - i. Binomial probability distribution is symmetric if $p = 0.5$
   - ii. Binomial probability distribution is skewed or asymmetric if $p \neq 0.5$
   - iii. Binomial probability distribution is positively skewed if $p < 0.5$
   - iv. Binomial probability distribution is negatively skewed if $p > 0.5$
3. The mean of binomial probability distribution is np.
4. The variance of binomial probability distribution is npq or $np(1-p)$.

9. **Hypergeometric Experiment**

A series of 'n' dependent trials, repeated without replacement and each having only two possible outcomes, success and failure, is called Hypergeometric Random Experiment.

10. **Properties of Hypergeometric Experiment**

Properties of Hypergeometric Experiment are as follows:

1. The two possible outcomes of a hypergeometric experiment are categorised as success and failure.
2. The experiment is performed a fixed number of 'n' trials.
3. Probability of success varies throughout the 'n' trials.
4. The successive trials are dependent.

11. **Hypergeometric Random Variable**

A random variable whose unpredictable possible values are determined by the outcomes of hypergeometric random experiment is called hypergeometric random variable.

12. Hypergeometric Probability Distribution

The probability distribution of hypergeometric random variable is called hypergeometric probability distribution.

Its probability mass function is as follows:

$$P(X = x) = \frac{{}^{k}C_{x}\,{}^{N-k}C_{n-x}}{{}^{N}C_{n}} \qquad \begin{array}{l} x = 0, 1, 2, 3, \ldots, n; \text{ if } n \le k \\ x = 0, 1, 2, 3, \ldots, k; \text{ if } k < n \end{array}$$

Where,
N = total number of items
k = number of items classified as success
$N - k$ = number of items classified as failure
x = number of success

13. Properties of Hypergeometric Probability Distribution

Properties of hypergeometric probability distribution are as follows:

1. There are three parameters of hypergeometric probability distribution, n, k and N.
2. The mean of hypergeometric probability distribution is $\frac{nk}{N}$
3. The variance of hypergeometric probability distribution is $\left(\frac{nk}{N}\right)\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$.

14. Poisson probability distribution

The Poisson random variable x with parameter μ has the following probability Mass function

$$P(X) = \frac{e^{-\mu}\mu^{x}}{x!} \qquad for\ x = 0, 1, 2, 3 \ldots$$

15. Characteristics of Poisson distribution:

in a Poisson process, with parameter μ, discrete events (successes) are generated in a continuous interval of time, length, are, space etc in the following manner.

(i) It is possible to divide the continuous interval of interest into many small non-overlapping sub intervals of equal length.

(ii) The probability that an event occurs in a given unit of time, length, area or volume is same for all the units of continuous interval.

(iii) For a subinterval the probability that exactly one success occurs is proportional to the length of the interval

16. Multinomial Distribution

The multinomial distribution can be looked upon as an extension of the binomial distribution, when there are more than two possible outcomes of each trial.
e.g. an item can be good, average or inferior.
Or A road accident may result in no injury, minor injury, major injury or fatal injury.

17. Multinomial Experiment

Multinomial experiment is the one that consists of n independent repeated trials resulting in any one of k possible outcomes such that k possible outcomes are mutually exclusive and exhaustive with probabilities for their occurrences in a single trial.

18. Conditions of a Multinomial Experiment

A multinomial experiment must satisfy the following conditions.

i. There are fixed number of n identical trials.
ii. Each trial has k possible outcomes that are mutually exclusive & exhaustive.
iii. The probability of i-th outcome pi remains constant and ∑Pi=1
iv. The trials are independent
v. The k dimensional random variable is the total number of times event occurs in n trials.

19. Multinomial Probability Mass Function

The multinomial probability distribution with parameter n and is specified by the probability mass function

$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k,) = \left(\frac{n!}{x_1!x_2!\dots x_k!}\right)(p_1)^{x_1}(p_2)^{x_2}\dots(p_k)^{x_k}$$

$$Mean = np_i$$

$$Var = np_i q_i$$

20. Negative Binomial Experiment

In the binomial experiment, the number of success varies and the number of trials is fixed
In "negative binomial" experiments number of success is fixed and number of trials varies to produce the fixed number of success.

21. Conditions of a Negative Binomial Experiment

A negative binomial experiment possess the following four conditions.

i. The experiment is repeated a variable number of times to obtain a fixed number of successes.
ii. Each trial has only two possible outcomes (Success and Failure).
iii. The probabilities of the two outcomes (Success and Failure) remain constant throughout the trials.
iv. The trials are independent

22. Negative Binomial Probability Distribution

When X denotes the number of trials to produce k successes in a negative binomial experiment, it is called negative binomial variable and its probability distribution is called negative binomial probability distribution

$$P(X=x)=p(x)=\left({}^{x-1}C_{k-1}\right)(p)^{k}(q)^{x-k} \quad for \quad X=k,k+1,k+2...$$

Where R.V X assumes a value x, on which the kth success occurs

This distribution is also called "Pascal Distribution" with 2 Parameters k & p

In negative binomial distribution mean < variance

$$Mean=\frac{kq}{p} \qquad Var=\frac{kq}{p^{2}}$$

23. Geometric Experiment

When an experiment consists of independent trials with probability of success p and the trials are repeated until the first success occurs, it is called geometric experiment.

24. Conditions of a Geometric Experiment

A geometric experiment has the following four conditions.

I. The experiment is repeated a variable number of times to obtain first success.

II. Each trial has only two possible outcomes (Success and Failure).

III. The probabilities of the two outcomes (Success and Failure) remain constant throughout the trials.

IV. The trials are independent.

25. Geometric Probability Mass Function

Let X denotes the number of trials needed for first success, then X is called geometric random variable and its probability distribution is called geometric probability distribution

$$P(X=x)=p(x)=(p)(q)^{x-1} \quad for \quad X=1,2,3,...,\infty$$

It is a special case of negative binomial distribution when k = 1

Since geometric r.v. represents how long one has to wait for a success, it is also called waiting time r.v.

This distribution is denoted by g(x; p) with one Parameter p.

$$Mean=\frac{1}{p} \qquad Var=\frac{q}{p^{2}}$$

Chapter # 9
Continuous Probability Distribution

1. The Uniform Distribution

The uniform probability distribution is perhaps the simplest distribution for a continuous random variable.

This distribution is rectangular in shape and is defined by minimum and maximum values.

The probability density function of a continuous random variable X uniformly distributed on the interval [a, b] is given by:

$$f(x) = \frac{1}{b-a} \qquad for \quad a \le x \le b$$

$$E(X) = \mu = \frac{a+b}{2}$$

$$Var(X) = \sigma^2 = \frac{(b-a)^2}{12}$$

2. Normal Distribution

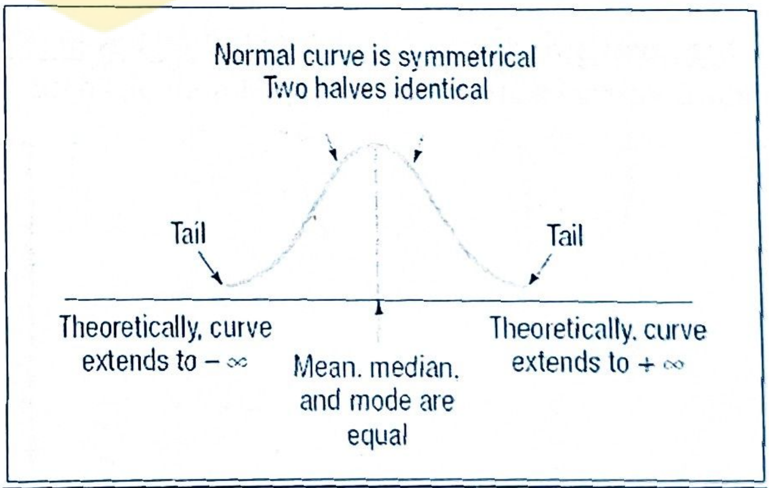
Normal distribution is a special case of binomial distribution when number of trials becomes sufficiently large, probability of success "p" and probability of failure "q" becomes approximately equal.

Binomial Distribution $\xrightarrow[p \to q \text{ or } p \to 0.5]{n \to \infty}$ Normal Distribution

The ordinate function and the probability density function of the normal distribution is as follows:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < +\infty$$

Where, μ = Mean of x (any real number) σ = Standard Deviation of x (any positive number only) π = 3.1416 e = 2.7183



3. **Importance of Normal Distribution**
   1. Many natural, psychological, educational and other practical variables like reading ability, introversion, job satisfaction, and memory are distributed approximately normally.
   2. Normal distribution is easy for mathematical statisticians to work with, and many kinds of statistical tests are based on normal distributions.
   3. It has one of the important properties called central theorem. Central theorem means relationship between shape of population distribution and shape of sampling distribution of mean.
   4. Marquis de Laplace proved the central limit theorem in 1810.
   5. The normal distribution is important because any linear transformation of a set of normal random variables also follow the normal distribution.

4. **Standard Normal random variable**
Standard normal random variable z is obtained by taking the deviation of normal r.v from its mean and then dividing by its standard deviation.

i.e., if $X \sim N(\mu,\sigma^2)$ then its standard variable is $Z = \dfrac{X-\mu}{\sigma}$.

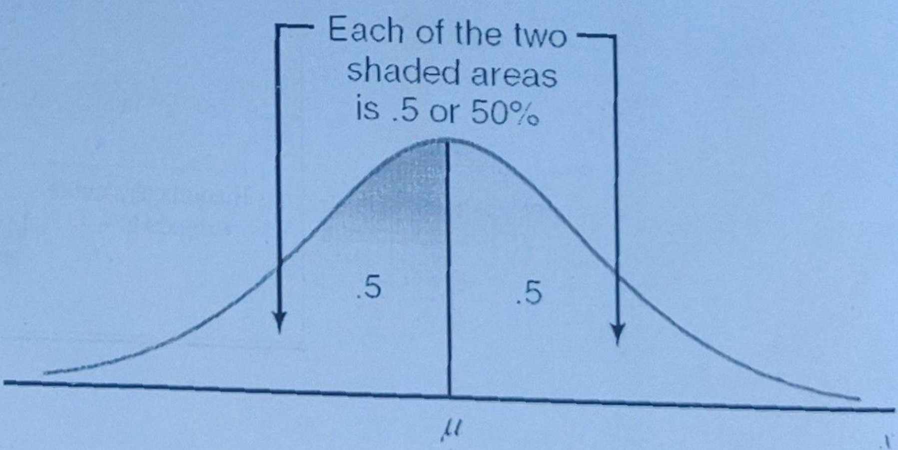
5. **Standard Normal Distribution**
If $X \sim N(\mu,\sigma^2)$ then by the reproductive property of normal distribution its standard variable Z also follows the normal distribution with mean 0 and variance 1.The probability density function of the standard normal distribution is as follows:

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \text{ for } -\infty < z < +\infty$$

Where, $\pi = 3.1416$
$e = 2.7183$

6. **When normal distribution becomes standard normal distribution?**
Normal distribution becomes standard normal distribution when μ = 0 and σ2 = 1

7. **At what point standard normal distribution has maximum ordinate?**
Standard normal distribution has maximum ordinate at Z = 0



Properties of Normal Distribution

Properties of Normal Distribution are as follows:

Type: It is a continuous probability distribution

Range: The normal random variable ranges from $-\infty$ to $+\infty$.

Shape: It is a bell shaped symmetrical and uni-modal distribution.

Averages: In normal distribution Mean=Median=Mode.

Quartiles: Lower & upper quartiles Q_1 and Q_3 are equidistant from mean.

$$Q_1 = \mu - 0.6745\sigma \quad \text{and} \quad Q_3 = \mu + 0.6745\sigma$$

Dispersion: $Q.D = \frac{2}{3}\sigma \text{ and } M.D = \frac{4}{5}\sigma$

Moments:

i. All odd order moments about mean are zero i.e, $\mu_1 = \mu_3 = \mu_5 = \ldots = 0$

ii. Second moment about mean is equal to variance. $\mu_2 = \sigma^2$

iii. Fourth moment about mean is $\mu_4 = 3\sigma^4$

Moment ratios:

i. Since normal distribution is symmetric and $\mu_3 = 0$ so $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 3$

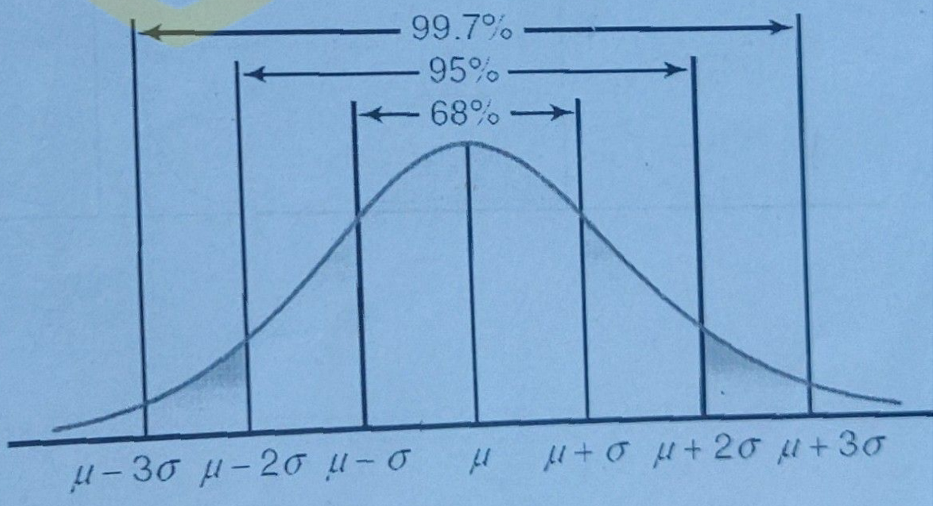
ii. Since normal distribution is Mesokurtic so the value of β_2 is always equal to 3.

i.e., $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{(\sigma^2)^2} = 3.$

Area: The total area under the normal curve is unity

Empirical Rule:

$$P(\mu \pm \sigma) = 68.27\% = 0.6827$$
$$P(\mu \pm 2\sigma) = 95.45\% = 0.9545$$
$$P(\mu \pm 3\sigma) = 99.73\% = 0.9973$$



Ordinate: Its maximum ordinate is at x = μ. $f(\mu)=\dfrac{1}{\sigma\sqrt{2\pi}}$

Asymptotic Curve: Its curve grows closer to x-axis but never touches it.
Points of inflexion: The points in the normal curve where it changes its concavity are called the points in inflection.
The points of inflexion for normal curve are $\mu-\sigma$ and $\mu+\sigma$.

2. **Reproductive property:**
   i. Any linear transformation of a set of normal random variables also follow the normal distribution.
   ii. If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ where X_1 and X_2 are two independent normal random variables, then the following conditions hold: $(X_1 \pm X_2) \sim N(\mu_1 \pm \mu_2, \sigma_1^2+\sigma_2^2)$

9. **What is the role of μ in normal distribution?**
It controls the location of the normal curve.

10. **What is the role of σ^2 in normal distribution?**
It controls the relative flatness of the normal curve.

11. **What happens if σ^2 increases?**
Normal curve becomes flatter.

12. **What happens if σ^2 decreases?**
Normal curve becomes sharply peaked.

13. **What happens if μ changes?**
There will be no effect on shape but it changes the center of the curve.